ASSIGNMENT #03

CHE301: ANALYTICAL CHEMISTRY & INSTRUMENTATION

Spring 2021

Total Marks: 10
Due date: 13-08-2021

VU Medical Zone Effort by Amaan Khan

Question No. 1

a) What is the difference between a Chromophore and an Auxochrome? Give one example of each. (6)

Chromophore	Auxochrome
Chromophore can be defined as the part of a molecule which is exposed to a visible light and it can absorb light and emit a certain light.	Auxochrome can defined as the group of specific atoms which contains functional group and it has ability to change the capacity to reflect the light from the Chromophore.
It always responsible for the color of the colorless compounds.	It is responsible for color intensity of the chromophores.
Example: Pale yellow color due to **Nitrobenzene**.	Example: Colorless **Benzene**.

b) Briefly discuss the Woodward-Fieser rule for Conjugated Dienes. (4)

A **diene** is either homoannular with both double bonds contained in one ring or heteroannular with two double bonds distributed between two rings.

Woodward-fieser can derived rules which attempt to predict the wavelength of the absorption maximum (λmax) in an ultraviolet and visible spectrum of a given compound for example conjugated carbonyl compounds.

Most of the absorption of a particular wavelength of light is depend upon the π-electron system of the molecule. The conjugation of the π-electron system within the molecule in which the higher the wavelength of light it can absorb. Robert Burns Woodward and Louis Fieser put down a set of rules which allows one to calculate the wavelength of maximum absorption (λmax) for a molecule empirically. These sets of rules to calculate the wavelength of maximum absorption or λmax of a compound in the ultraviolet-visible spectrum, based empirically have been called the Woodward-Fieser rules or Woodward's-rules.

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